Experience and re-experience without boundaries in time and space: the extension of a practical tool

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Abstract: "…you feel like you´re in the exhibit. It´s surreal. It´s awesome. It´s the future", an American architect this summer blogs when, in the CAD[1] model he downloaded from the museum's website, virtually having visited the Carl Larsson exhibition on show in Stockholm.[2] This quotation expressively highlights the topic of this paper in which the focus is on questions concerning adding value in curatorial practice, especially concerning public and museological aspects, when using 3D modeling as a tool. The aim is also to discuss the analytical implications in the light of the Digital history field. How could an architectural CAD program improve the installation planning process? How could this practical tool be used for museological analysis and research and what kind of representational aspects will then be highlighted; would it enable visual experience in virtual space? These rather diverse questions, the first offspring from the curatorial practical and pragmatic everyday work at a large art museum, and the second and third from a museological and digital history analytical perspective, will function as departure for reflections on digital tools in contemporary museum work and museum studies. The main concern is on the spatial and visual aspects connected to visualizations of exhibitions and questions interrelated to this topic, leaving all questions concerning the ongoing digitalization of the collections and other main documents aside.

Keywords: 3D modeling, curatorial practice, museology, visualization, representation, experience visuals

Background

3D digital visualizations are since the last two decades of the twentieth century a well established tool within architectural circles but not at all broadly disseminated in the museum world. The museum sector seems to prefer traditional physical exhibition design drawings and physical 3D models and at Nationalmuseum in Stockholm, Sweden the use of CAD has often been forgoing by traditional methods. This situation could be explained by the fact that most museums, even the national institutions, lack internal competence in the digital field and some exhibition designers seems to still prefer pen and paper. Ten years ago, and still ongoing, the museum directors at Nationalmuseum discussed different ways of improve the exhibition planning process in order to reduce the time needed for installation. The arguments were for the benefit of the public and economical reasons and from this management requests some members of the technical staff took an initiative to investigate if the use of 3D modeling could improve efficiency and support for the art handlers.

In the beginning, in year 2003, this method was used as a compliment to the exhibition drawings, giving possibilities for curatorial changes in details before the installation began and still being able to through the CAD produce very accurate installation plans to the art handlers. The CAD programming was added quite late in the overall process, when the design of the display first had been presented by the exhibition designer and accepted by the curator. Then the CAD work began with adding the image files, following the sketches and sometimes the physical model done by the exhibition designer, consisting exact measures of the objects and their spatial relations in the galleries. It was not obvious for the curators in what way they could benefit of using the digital tool or why it should be used by them at all. Could it really replace the good old physical model and story line was a common question, but even though they did not actively use the digital model during the planning process, they had to approve the display in the CAD model before it was sent to the art handlers. They in turn were impressed; now they got clear and detailed drawings that had passed the curatorial board and been

[1] CAD, Computer Aided Design

[2] http://www.shoegnome.com/2013/07/16/tour-of-exhibition-via-bimx/, http://www.nationalmuseum.se/sv/English-startpage/Exhibitions/Past-exhibitions-/Carl-Larsson--Friends-and-Enemies/A-Tour-of-the-Exhibition-in-3D/

approved, which were of great help in the overall planning and reduced last minutes changes when in the middle of installation. It soon became obvious that the installation process had become more efficient. The time needed for preparation and installation had been reduced with days up to a week.

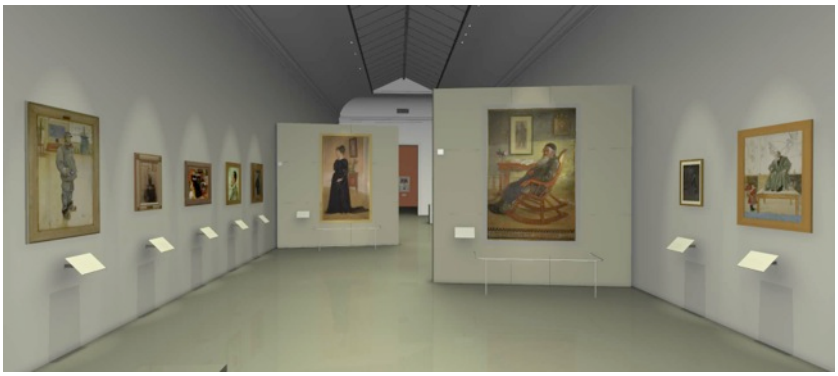


Image 1: CAD image from the Carl Larsson *exhibition at The Royal Academy of Art 2013*

Ten years have passed and this method, which awoke from pure curiosity and a will to explore the possibilities of using new technology for improving visualization of the exhibition display, is implemented as a routine in the exhibition planning process. By now it is not only a practical tool for art handling, it has also slowly grown to be a curatorial aid early in the exhibition planning; a way to structure the narrative in the digital environment; sometimes without first having tested the display in a physical model. Almost all hangings and re-arrangements of the collections are currently planned and visualized in CAD. The exhibition model is also published on the museum web site enabling virtually exhibition visits without boundaries in time and space.

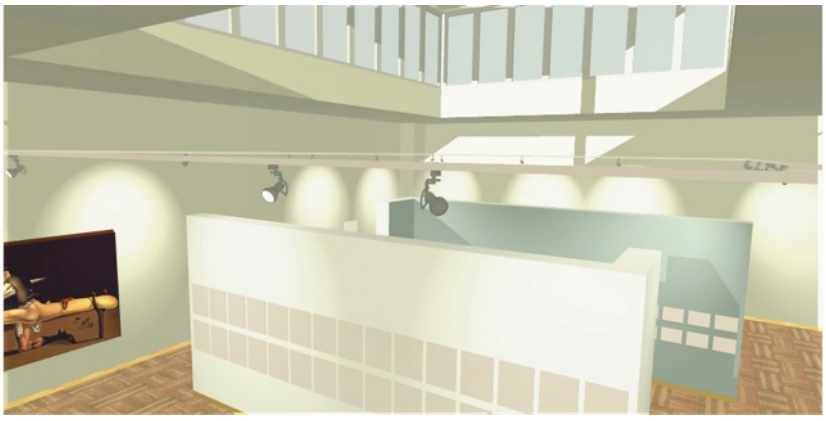


Image 2: CAD image from the exhibition From El Greco to Dali at Nationalmuseum 2003

A curatorial tool

From a curatorial point of view the benefits of being able to visualize the spatial relations within the galleries especially when putting objects of different materials, and different size on display has been of great importance; the possibility to move around, to go near or step back in front of a painting or a group of objects on display; to spatially experience the route a spectator might choose or to ensure that there will be space enough for a group of visitors joining a guided tour are other curatorial aspects which you might experience and try out in the CAD model.

The digital display also has collaborative potentials as everyone with an ordinary computer could get accurate information of the display. Recently in the exhibition *Passions* at Nationalmuseum this way of collaborating was tried out in the discussions concerning the display of the video installations *The Passions* by the American artist Bill Viola. The Gallery in Los Angeles at first insisted to send a courier to monitor the installation; requesting first class flight tickets and five nights stay in Stockholm, but after having seen the model and talked to our CAD designer, they found their presence needless due to the information of the display they had got from the model.



Image 3: Interior from the exhibition Passions at Nationalmuseum 2012

From this practical developments with the CAD followed a short but interesting experimental step into another virtual space. When hearing about the Swedish Institute plans to establish The House of Sweden, build 2006 by the architect Gert Wingårdh, and housing the Swedish Embassy in Washington D.C., in the digital world Second Life, Nationalmuseum proposed to put some objects from the collections on display in the digital building; an extension of the dissemination of the collections that has been part of the Museums activity since early twentieth century. The idea was realized in collaboration with the Swedish Institute and later on the museum even offered two traditional guided tours for a bunch of interested avatars.

An analytical museological tool

In recent years there is a tendency to study museum history from a visual culture perspective rather than the sociological and political focus that dominated the field during the last decade of the twentieth century. In *Spaces of Experience. Art gallery Interiors from 1800 to 2000*, Charlotte Klonk investigates what kind of experiences western art museums have generated through different display principles and in relation to the ideals of a certain period. She finds the museum, as a good example of places in-between the public and the private sphere, very useful in all studies concerning the "*cultural history of experience*".[3] The experience is due to social and historical forces and according to Klonk the museum display expresses the consciousness of the curators concerning what kind of experience they want to give to the public.[4]

In accordance to Klonk and her idea of the representational function of art exhibitions one aim in this paper is to investigate the analytical potentials when using the CAD for reconstruction of historical exhibitions. What kind of questions does a historical reconstruction awake? Does it offer any analytical added value comparative to the study of a black and white photography? This kind of questions have been of interest for me during the last couple of years and in consequence to an interest in historical exhibitions and currently in connection to my PhD project which deals with the temporary exhibitions at Nationalmuseum during the period 1866 – 2012, with special focus on the relation between the exhibition practice, the art historical discourse and the socio cultural ideas of the same period.

[3] Charlotte Klonk, Spaces of Experience. *Art Gallery Interiors from 1800 to 2000,* Yale University Press New Haven & London 2009, p.3.
[4] Ibid, p. 9.



Image 4: Black and white image from *Sveriges Nationalmuseum*, the Swedish Gallery before the re-hanging 1906.

The case study chosen for this investigation is a digital reconstruction of the collection display from 1906.Then the picture guidance *Sveriges nationalmuseum,* written by the author Anders Hasselgren, was published.[5] It is a lively and, with more than 400 images, detailed description based on a fictional tour through all the departments in the museum; in this period not only a fine art museum, but as a National museum, and like many of the European museums, born in the age of nationalism during the 19th century, Nationalmuseum was a museum of the nation. As a National Museum, the museum had to host the historical collections, the national antiquities, the armory collections, the coin collection, the sculptures and the collections of applied art and finally on top floor the collections of fine art which in fact meant that it already from the beginning was perceived as to overcrowd. The Venetian renaissance façade representing classical references was in the entrance hall combined with the Swedish mythology, the giant sculptures by Fogelberg: Oden, Tor and Balder. The collections were arranged according to the common and hierarchical ordering of museum collections in the late 19th century. At the bottom were the archeological and historical collections, in the middle floor the armory and coin collection together with the collections of applied art and then at last, at the top of the temple, the collections of fine art surrounded by the plaster casts of the antiques placed along the wall surrounding the banister. Before reaching the top floor the visitors also had a possibility to gain some acquaintance with another famous antique; a plaster cast of the Parthenon frieze which went along the walls of the staircase.

The fine art collections were presented in a traditional manor in schools dominated by the Flemish and Dutch 17th century, the French 18th century, the Italian school and the Swedish 18th and 19th century. Looking from a museum perspective Nationalmuseum definitively represented the modern and successful nation Sweden, although not a great power nation any longer but instead bearer of a cultural and national modern spirit fully comparable to its European neighbors.

The guidance combination of images and text provides an interesting source for further analysis and it even comprises the re-arrangements of parts of the collection which took place later the same year. Then there was a major change in the display of the Swedish and Nordic schools and this change is also recorded in the guidance. What does this re-ordering tells us and could a reconstruction of the display/visualization of the exhibition reveal the curatorial motives behind the installation in a way a two dimensional black and white photo could not?

[5] Hasselgren, Andreas, *Sveriges nationalmuseum, dess konstskatter och historiska samlingar i bilder*. Stockholm 1906.



Image 5: CAD reconstruction 1906 display

This example shows the display of the Swedish nineteenth century paintings. It is a tight, three leveled salon display and it is quite easy to grasp that aspects as size and topic have been considered in the overall design; following the principles for art display during the late nineteenth century. In the guidance the visitors' senses are triggered by descriptions as: violent, lively, sensuous, brutal, light, cold, powerful, impressive etc. together with information concerning the artistic qualities and sometimes a brief biographical note. But how is this visualized in the exhibition? The black and white photo reveals and the reconstruction confirms a curatorial notion based on the dominant art historical narrative in this period when the academic tradition, in this gallery represented by Swedish historical and Nordic mythological topics, Swedish landscape and portraits, still was regarded as representative for the Swedish national spirit.

When working on the reconstruction of the re-arrangement of the 1906 display something odd and peculiar caught the attention. Why were *Young Boy Peeling a Pear* by Édouard Manet and *Talking in the Twilight* by another French painter, Lucien Simon, centrally displayed in the gallery destined for the Swedish and Nordic schools from the same period? Of course there could be pure practical and pragmatic answer to the question due to the rather poor representation of late nineteenth century French paintings in the Nationalmuseum collection by this time. But even though there could have been some pragmatic curatorial motive behind this display it still has a representational function raising questions concerning interpretation and experience. How did it fit the art historical narrative built on nationalities and schools? The two French paintings are centrally located on the wall which indicates a curatorial decision based on art historical notions concerning modernity and quality, but there is also an interesting diagonal line which combines the three portraits where the Manet painting is the middle one. It is possible to sense the grouping based on size and localization on the wall but also the overall spatial relations between the two opposite walls. Color is another important aspect brought to light in the reconstruction but the effects of lightening has so far not been able to explore and analyze.

As there are few resources actually describing an experience of the gallery interior it is necessary to analyze the curatorial documentation, and other comparative material of interest. Klonk studies the scientific developments concerning perception and the public space of fairs – the choreography of desire. She is looking for connections avoiding a parallel study, rather focusing on experiences shared by individuals with certain interests and needs. She is more influenced by Bourdieu but unlike Bourdieu she is not primarily interested in viewing habits but in what kind of viewing perspectives museums have transmitted to the public.[6]

In this case study the main focus has been on the visualization with an intention to investigate what kind of information a CAD reconstruction could offer. So far the 3D model has offered an immersive sense of the overall spatial relations in the gallery, but also the curatorial concerns based on colors, balance and variation. Who then is the ideal visitor in this sense? The ideal visitors to the Swedish gallery before the re-arrangement were probably thought to be men and women, but also families, from the expanding bourgeois or middle class, with a pre-conceived knowledge of the historical and/or Nordic mythological episodes depicted but also susceptible of the direct and sensual mood expressed

[6] Ibid, p. 11.

in the large amount of academic landscape paintings; an aesthetic art consumer striving for pleasure. The re-arrangement is not radically different except for the two French paintings, it is very much governed by similar esthetical principles; the overall impression is more spacious and the Nordic paintings are gathered on the same wall. But although small, the difference indicates a curatorial sense of modernity. The reconstruction also supports an analysis from the point of view of the spectator. In the model it is possible to reenact a performance in the gallery space, the movements and direction of gaze, this performance reveals the curatorial aim to through variation, groupings and perspective alert the visual perception and the senses; an experience built on similar perceptual principles as the visual media of today.

Digital history: representation, authenticity and experience

In a Nordic PhD student course in Digital History at Umeå University during the autumn 2013 one of the main questions discussed was how do digital tools and digital material change the study of the past? What kind of new demands have future historical studies to face when dealing with digital sources? [7]

Following the digital boom and during the beginning of the twentieth first century many critical voices have discussed and reflected upon the impact of digital development in museums. In the light of digital humanities and aspects as representation, authenticity and experience the museums social function in future society have been questioned. The participatory museum and other public aspects have been one of the main issues discussed by museum professionals and scholars; a debate often expressing the hopes and fears of the digital turn in the museum world. [8]

In an online interchange course published in an article in *The Journal of American History* 2008 (JAH): *Interchange. The promise of Digital History, where experts within the digital humanities and digital history* prominent scholars debated many different questions related to the academic studies of the digital field. This discussion also touched upon some museological and analytical perspectives which revealed the most common objections and fears when talking and using digital tools in museum practice. First there was a question of real experience, an expectation that the audience wants to face reality, to gain an authentic experience which preferably occurs in relation to a museum object.[9]



Image 6: Black and white image from Sveriges Nationalmuseum, the Swedish Gallery, after the re-hanging 1906

[7] http://www.digitalhistorian.net/umedh/
[8] f.e. discussed at the Nodem konference in Copenhagen November 24-26 2010, with papers published in *Nordisk Museologi, 2011:1,* Tema: Digital Museologi; Simon, Nina, *The participatory Museum,* California 2010.
[9] http://www.journalofamericanhistory.org/issues/952/interchange/index.html

In this sense the artifacts are decisive but according to one of the debaters, Patrik Gallagher, leader in Exhibition design there is possible to create a narrative and an experience without artifacts. For him the ideal digital exhibition or display functions both as a teaser and a following up, which would encourage people to visit the museum, preferably not alone, and then afterwards use it for immersion. He also argued that the sense of reality in digital exhibitions could be elaborated even further with adding animations/moving pictures and the voices of the persons involved in the presentation or historical moment.[10]

The crucial question is perhaps what kind of experience you could have in a virtual environment. Gallagher stated that there are no big differences but he finds the loss of scale the most significant in the digital space.

Amy Murrell Taylor, associate professor of history at the State University of New York, expressed the second fear. Rhetorically she asked if digital representations would replace traditional history documentation and would museum web sites be an alternative to actual visits to the museum. Quickly answering, she concluded that the sensory and physical experience felt when meeting old artifacts will continue to attract people.[11]

According to Steven Mintz, director of the Graduate School of Arts and Sciences Teaching Center at Columbia University, the hands-on offering is not considered as a being interactive. For him true interactivity occurs on an intellectual and communicative level and he found this possibility within the virtual museums:

> Ironically, virtual museums may be more capable of this than their physical counterparts. Virtual exhibitions can allow users to magnify objects for closer scrutiny. These exhibits can offer expert commentary and links to glossaries and other reference material. In addition, there is no issue of "flow"; one can spend as much time as one wishes with an object.[12]

This brief summary of the section of the scholarly discussion in JAH that focused on digital aspects in museums is of course only one small piece of the much larger discursive field within digital humanities. But it clearly reveals the questions and fears still in focus when reflecting and debating digital aspects; questions concerning authenticity, reality and experience. At the Nodem Conference in Copenhagen 2010 these aspects were also discussed by Dr. Chiel van den Akker.

Dr. Akker identifies three different ways of approaching and experience a cultural heritage object. It could be in a traditional original environment; in a preserving environment as f. e. museums or in a representational environment. According to Akker a virtual representation; a digital object could be regarded as representative of a tradition and in that aspect also a documentation of the history. He argues that digital objects are representations; images representing the original object and in a digital environment all collections are documents. Closely linked to the question of cultural heritage and representation is the concept of authenticity. Following the definition Walter Benjamin elaborated in *The Work of Art in the Age of Mechanical Reproduction* Akker concludes that there are two senses of authenticity. The first is related to a unique object´s accumulated history and the second sense of authenticity is related to documentation. Reproductions, e.g. digital images, do not have a unique status or a fixed position in time and space and therefore they lack authenticity in the first sense but virtual representations are documentations of unique object´s; a part of their history, and therefore related to the second sense of authenticity.[13]

The CAD exhibition models are in this respect documents, representations of an intentional historical display, in a similar way as the physical 3D models. But as digital born files and representative documents of a museum history they should not only be a residue product from the exhibition planning process but a source for future investigation enabling detailed analysis of both curatorial aspects and the experience accomplished by the visual and spatial relations.

In an American survey from 1998 the respondents expressed a wish to at a larger extent be part of the construction of history offered by, for example, museums. It was not primarily due to lack of trust in the academic expertise but rather a preference for choosing path. The authors of the survey conclusively

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[10] Ibid.
[11] Ibid.
[12] Ibid.
[13] Van den Akker, Chiel, Tradition (re)visited: From Place to Presence. *Nordisk Museologi 2011:1,* pp.107–110.

suggested that museums professionals should open up for a wider participation, like the one made possible through different web 2.0 collaborations.[14]

The digital visual and spatial developments have been explosive giving new opportunities to public participation and experience which have been of great interest for museums. But yet there is hardly possible to talk about a digital revolution in the sense of a total re-orientation in museum practice. The traditional way of one-way communicating is still dominating and museums tend to keep the power of interpretation and communicating; the digital public support offers a variety of predetermined choices, but the audience is still directed by the authoritarian curator.[15]

But, on the other hand, as the main focus in museological studies seems to have shifted from the social and political to the visual and perceptional aspects, there is also a new interest in the experience as analytical category; a turn from an interest in the museum as part of the national project to the spectator; the individual. This alternative perspective may be not radically different, but it indicates a changing focus; from the museum body, the mute art object and the passive culture consumer to the spectator as a creative force in this other story. With reference to the introductory quotation the question still remain: how could digital experience contribute to future museums perspectives?

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